Lecture 14

Thursday, 23 February 2023 1:35 PM

$$i = -F^2 \nabla\phi \sum z_i^2 u_i c_j - F \sum z_j D_j \nabla c_j + F v \underbrace{\sum z_j c_j}_{0}$$; Convection does not constitute a current

For a well-mixed $\nabla c_j = 0 \Rightarrow i = -F \nabla\phi \sum z_j^2 u_j c_j$; $\kappa = F^2 \sum z_j^2 u_j c$

$\therefore \boxed{i = -\kappa \nabla\phi}$ ← Ohm's law

Fluid flow →

High pot. high conc. — ⊕ → Conv. diff. mig. ; mig. ⊖ → Conv. diff. — Low potential Low conc.

Even in the case of migration only, ions move differently depending on chemical identity, conc, charge → carry diff. currents

↳ We can characterize the fractions of current carried by a species as the transference #

t_i : $i = -F^2 z_i^2 u_i c_i \nabla\phi = \underbrace{\frac{z_i^2 u_i c_i}{\sum_i z_j^2 u_j c_j}}_{t_i} i$; i = individual species

→ When modelling reaction - conv - diff - mig. } We use material balances that adhere to concepts of mass + charge

$$\frac{\partial c_j}{\partial t} = -\nabla \cdot \bar{N}_j + \cancel{R_j}$$ often 0 but ...

→ precipitation

→ $H^+ + OH^- \rightleftharpoons H_2O$

↳ any buffer reaction

ex) bicarbonate

} Any homogeneous reaction

SEI → faradaic process actually (heterogeneous)

$$\nabla \cdot N_j = \nabla \cdot \left(-z_j u_j F c_j \nabla\phi - D_j \nabla c_j + c_j \bar{v}\right)$$

$$= \nabla \cdot \left(z_j u_j F c_j \nabla\phi - D_j \nabla c_j\right) + \nabla c_j \cdot \bar{v}$$ Nernst-Einstein $u_j = \frac{D_j}{RT}$

$$\boxed{\nabla \cdot N_j = -D_j \left(\frac{zF}{RT} \nabla \cdot (c_j \nabla\phi) + \nabla^2 c_j\right) + \nabla c_j \cdot \bar{v}}$$

↳ We can write one of these eqn. for each species in our system (i.e. n eqs for n species)

↳ But we have (n+1) variables, considering ϕ

↳ Final equation → comes from electroneutrality

↳ This forces the electric field to take on whatever value satisfies this equation

$$\frac{\partial}{\partial t} \underbrace{F \sum z_j c_j}_{0} = -\nabla \cdot F \sum z_j \bar{N}_j$$ [$\frac{\partial c_j}{\partial t}$

definition of current (flux of charged species)

$$\boxed{-\nabla \cdot i = 0}$$

↳ Charge is conserved (divergence is zero) → since charge is carried by physical particles (& these are conserved)

Since $i = -\kappa \nabla\phi$

$$-\nabla \cdot i = -\nabla \cdot (\kappa \nabla\phi) = 0$$

$$\boxed{\nabla^2 \phi = 0}$$ → Laplace eq. (Valid for no conc. gradients)

These n+1 (flux + electroneutrality) give system of partial differential equations, need BC's:

↳ BC @ (insulating surface): normal comp. of flux is zero for all species

$\bar{n} \cdot \bar{N}_j = 0$ (n of these BC's + electroneutrality)

$\bar{n} \cdot \bar{i} = 0$

↳ BC @ (electrode surfaces): * Faraday's law as flux

↳ Flux of species @ electrode is proportional to current density

$\therefore \bar{n} \cdot \bar{N}_j = \frac{-S_j}{nF} i_n$ ← defined by B-V (or other relevant kinetic expression)

η is at electrode surface (must account for potential in electrolyte formed by N_j)

$\eta = (V - \phi) - (U - \phi^0)$

- V: applied potential
- ϕ: electrolyte potential
- U: electrolyte potential @ equil. ($i = 0$, $\phi^0 = 0$)
- U: electrode equil. potential

$\eta = V - \phi - U$ ← Thermo

- V: user-defined
- ϕ: variable to solve for PDE ($\nabla^2\phi = 0$ → or other forms)

↳ i_0 will also depend on c_j^{surf} → these are also solved for @ surface

Limiting cases for BC's:

↳ Limiting current density: @ large η, r_{rxn} may be fast that $c_{ion} = 0$ @ surface.
in this case current density @ electrode is defined completely by N_j

↳ Chemically limited rxn rate: @ slow rxn rates; $c_j^{surf} = c_j^{bulk}$ and we can neglect conc. gradients

↳ our only variable is ϕ, defined by $\nabla^2\phi = 0$

* and if ohmic resistance to current flow is much larger than charge transfer resistance (eg: poorly conducting electrode), then $V = \phi$ @ electrode surface

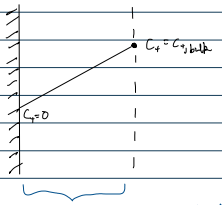
↳ Slow charge transfer rate: charge transfer resistance is much larger than ohmic resistance in solution, then we have uniform current density

$i_n = \frac{I}{\text{area}}$ @ all positions

(ohmic resistance: poorly conducting electrolyte)

Transport of Binary Electrolyte

↳ M^+X^- of a single salt species is present (migration is most important)
eg: Cu plating @ electrode where $C_{Cu^{2+}} = 0$ @ surface (fast rxn)



diffusion BL ($\delta \sim 10\,\mu m$) → Fick's law

$N_+ = -D\frac{(C_{+,bulk} - C_s)}{\delta}$